# Objectives, evaluation, and acceptance criteria

> **Reading note.** This is an explanatory reconstruction of Jiaxuan Zou's literature synthesis. Published experiments belong to the cited teams. Numbered protocols and illustrative matrices are our planning synthesis unless explicitly attributed.

The blog's §2 asks what decision the experiment must support before choosing a grid. A recipe that predicts loss accurately can still be less competitive than another recipe. Conversely, the lowest observed loss is not sufficient evidence that its advantage transfers to the target scale.

## Choose the decision

| Decision | Estimate | Evidence required |
|---|---|---|
| Forecast a fixed recipe | Target loss and interval | A fixed recipe plus untouched larger runs |
| Allocate compute | Best $N$, $D$ and regret | Independent variation in size and tokens |
| Compare recipes | Paired loss or capability difference | Comparable tuning and multiple scales |
| Transfer hyperparameters | Near-optimal settings at target | Sweeps across size and horizon |

Freeze evaluation data, weights, prompts, decoding, and scoring versions. Use continuous metrics when small models are at the accuracy floor. Bits per byte is $\mathrm{BPB} = \mathrm{NLL}/(n_{\mathrm{bytes}} \ln 2)$, where NLL is summed negative log-likelihood in nats and bytes are measured on the scored text. Prompt masking and answer-only scoring must be stated.

## Put thresholds in decision units

For $L(C) = E + AC^{-\gamma}$, with positive $A$, $\gamma$, a model whose loss is lower by a small positive $\delta$ has an approximate equivalent-compute advantage

$$\frac{C_{\mathrm{equiv}}}{C} \approx \exp\left[\frac{\delta}{\gamma(L-E)}\right].$$

This is a local approximation and becomes sensitive to the fitted floor $E$. Report raw loss differences alongside it. For two recipes, compute intervals on paired differences rather than treating all noise as independent.

1. Set a minimum meaningful gain and prediction-error tolerance before inspecting holdouts.
2. Estimate seed and evaluation noise in pilots.
3. Define capability floors and numerical stability requirements.
4. If post-training is part of delivery, include a small standardized post-training check.
5. Report an inconclusive result when uncertainty spans the acceptance boundary.

**Deliverable:** a frozen decision rule with metrics, tolerance, confidence procedure, and post-training checks.

**Sources:** blog §2; OLMo team, 2025, OLMo 3 §3.3; Choshen et al., 2025.

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